

Question 6

(a) Polymer-based injectable modified release systems come in a range of formats. One format is a polymer microsphere. Describe in point form the advantages and disadvantages of polymer microspheres as a dose form and what each means for the patient. (5 marks) (b) A rival technology to microspheres is 'in situ gelling' polymer systems. In point form briefly describe the attributes of these systems. Is there a clinical benefit to the patient for these systems over microspheres? Why do you think they are of interest? (5 marks)

Marking Rubric for Question 6

(a) Polymer-based injectable modified release systems come in a range of formats. One format is a polymer microsphere. Describe in point form the advantages and disadvantages of polymer microspheres as a dose form and what each means for the patient. (5 marks)

(1 mark each point up to 5)

Advantages

- May be biodegradable - do not need removal
- Provides a longer duration than lipid systems, reducing frequency of administration
- Injectable system, does not require surgical implantation
- Aesthetics – no physical lump where the implant is injected

Disadvantages

- Limited to non-acid labile drugs – patient may need alternative system in this case
- Need skilled health worker to administer – may exclude remote patients
- Complex manufacture - expensive

(b) A rival technology to microspheres is 'in situ gelling' polymer systems. In point form briefly describe the attributes of these systems. Is there a clinical benefit to the patient for these systems over microspheres? Why do you think they are of interest?

(5 marks)

- Systems contain a solvent, or polymers that are temperature sensitive
- Generally prepared using a biodegradable polymer
- drug must be soluble with polymer in solvent
- on administration solvent diffuses away

-drug released by degradation and erosion of the polymer matrix

-No clinical benefit, may be cheaper,

-easier to manufacture, easier to sterilise

- Avoids needle blocking

Question 8

Consider the following five statements, answering whether you (1) agree or disagree and (2) provide a brief reason for your answer. (a) Lyophilization is the only option for preparation of peptides for injectable administration. (2 marks) (b) Protein structure is more complex than peptides, meaning instability is often a greater issue. (2 marks) (c) Antibody-drug conjugates use the natural distribution of a drug to get the antibody to bind to its target on cells to deliver a therapeutic effect. (2 marks) (d) Proteins and peptides are not light sensitive, so packaging to protect the drug from light is never used. (2 marks) (e) Co-solvents are used to dissolve peptides to enable safe injection. (2 marks)

Marking Rubric for Question 8

Consider the following five statements, answering whether you (1) agree or disagree and (2) provide a brief reason for your answer.

(a) Lyophilization is the only option for preparation of peptides for injectable administration. (2 marks)

- One mark for disagree,
- one mark for not necessary if drug is stable in solution

(b) Protein structure is more complex than peptides, meaning instability is often a greater issue. (2 marks)

- One mark for agree,
- one mark for either hydrophobic parts of proteins can associate and stick to surfaces, or the tertiary structure may be disrupted by unwanted chemical reactions

(c) Antibody-drug conjugates use the natural distribution of a drug to get the antibody to bind to its target on cells to deliver a therapeutic effect. (2 marks)

- One mark for disagree,
- the antibody directs the drug to the target not the other way around

(d) Proteins and peptides are not light sensitive, so packaging to protect the drug from light is never used. (2 marks)

- One mark for disagree,
- One mark for mentioning amber vials

(e) Co-solvents are used to dissolve peptides to enable safe injection. (2 marks)

- One mark for agree,
- one mark for saying either: unsafe if proteins/peptides not completely dissolved, or injectable co-solvents are OK to use in small amounts